Signal Detection Experiment
PSY310 - Lab in Psychology
Lab Report

11th September 2025

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GitHub Link:
https://github.com/aryanm4/PSY310_LabInPsychology/tree/Signal-Detection

Introduction

Signal detection procedures are effective methods to measure one's ability to identify the presence of a specific stimuli and distinguish it from others. These methods are based on the Signal Detection Theory, which assesses people's ability to identify a stimuli and differentiate from when it is not present, such as detecting a specific sound amidst background noise.[1] These methods divide participant responses into four distinct categories: hit, miss, false alarm, and correct rejection.[2] A hit is when the participant correctly identifies the presence of the signal; a miss is when the participant fails to identify the present signal; a false alarm is when the participant wrongly identifies a signal which is absent; and a correct rejection is when the participant correctly identifies the absence of the signal.[3]

[1] Sherman, 2024
[2] Wickens & Flach, 1988
[3] Onifade, 2021

Method

Participant
The participant is an 18 year old male undergraduate student at Ahmedabad University. Full consent was provided and he was informed of the trial's objective, to determine whether the grating was tilted or not.

Materials and Procedure
This experiment was created on a desktop computer with a resolution of 1920x1200 pixels using PsychoPy-2025.1.1. The procedure assessed the participant's ability to identify whether the grating was tilted or straight. 100 trials of the experiment were conducted.
The trial began with a cross-shaped fixation presented for 500 milliseconds followed by the stimuli, a sinusoidal grating with a Gaussian mask (Figure 1), presented for 300 milliseconds. A code was added so that the grating was randomly tilted: either to the left or right; or completely straight with no tilt. The intensity of the tilt was randomly assigned between -5 degrees and -1 degree, or 1 degree to 5 degrees, when the grating was tilted, and 0 degrees when the grating was straight. This random assignment is shown in Figure 2. The participant was instructed to press the 'up' arrow key if the grating was straight with no tilt, and the 'down' arrow key if it had a tilt. The responses were collected on PsychoPy in an Excel spreadsheet, which were then analysed to determine the accuracy.

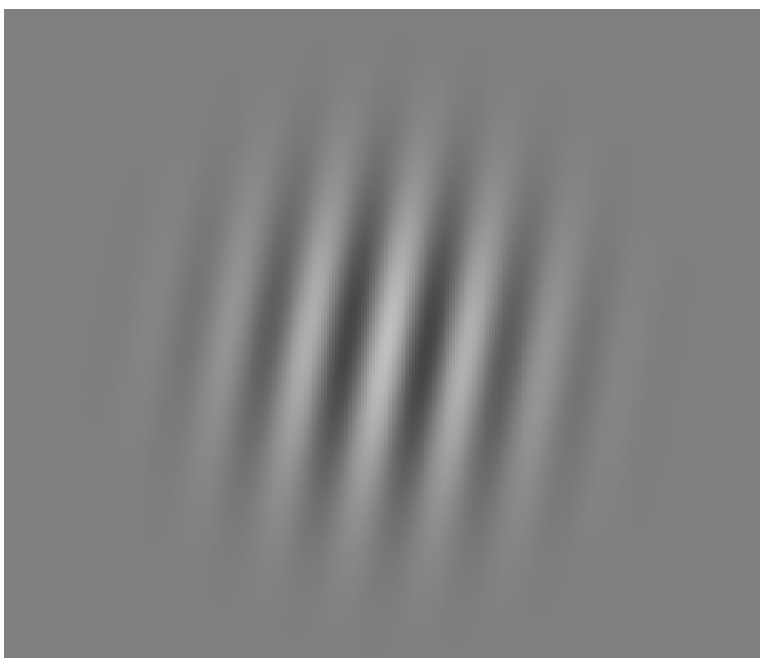

Figure 1 - Sinusoidal grating with a Gaussian mask, tilted 5 degrees rightward.

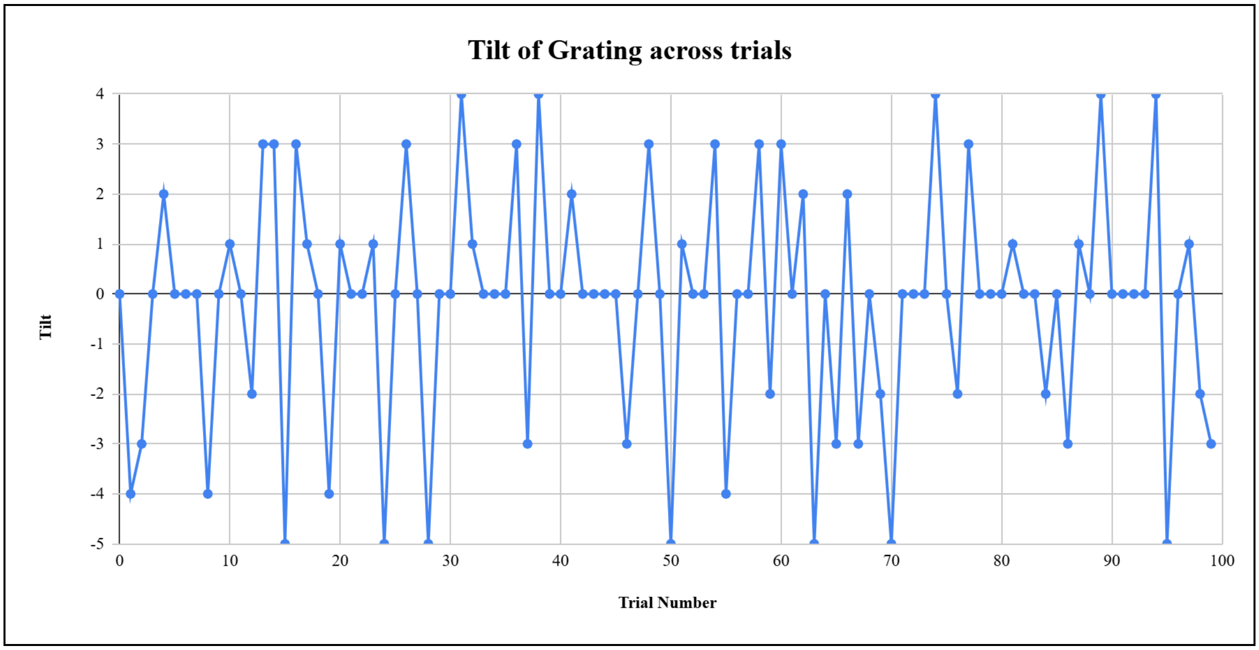

Figure 2 - The changes in tilt intensity from the first to the hundredth trial.

Results

The participant responded with a 78% accuracy across 100 trials, as shown in Figure 3. The graphs suggest that the participant responds inaccurately, i.e. a miss where there is a tilt but the participant responds with the ‘up’ arrow key, when the tilt of the grating is lower, such as 1 or 2 degrees, indicating a moderate level of signal detection ability.

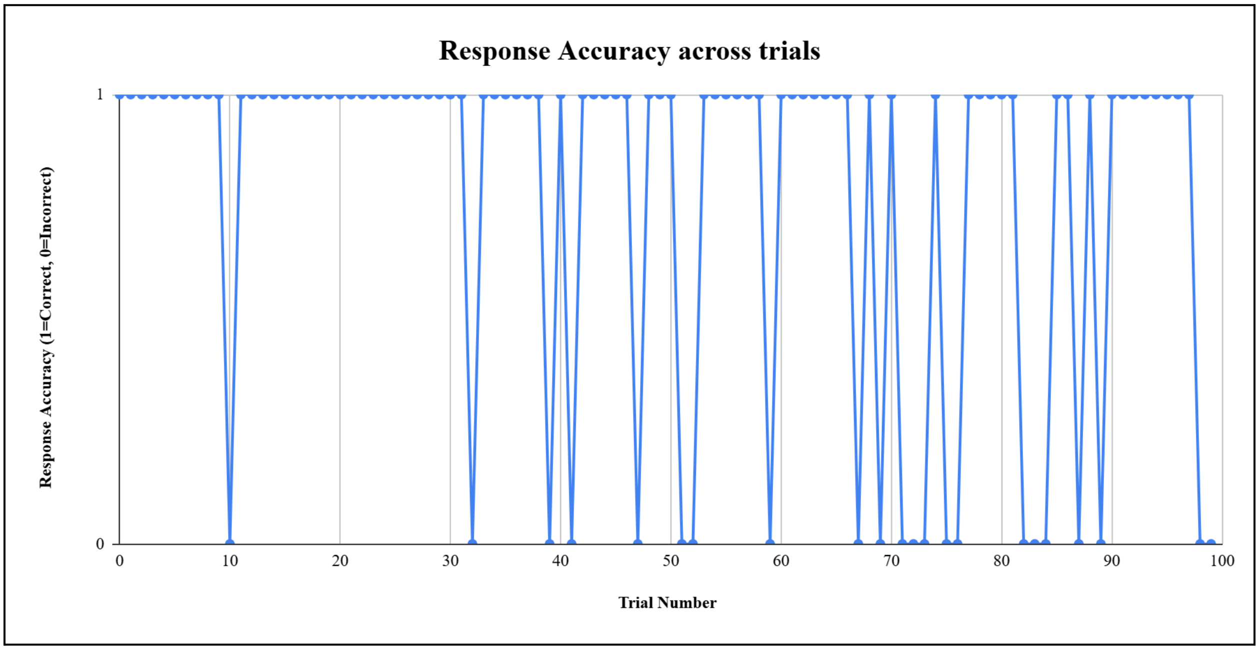

Figure 3 - Participant’s accuracy over the 100 trials.

Discussion

After analysing the participant's responses, a 78% accuracy was found over 100 trials. This indicates a moderate level of signal detection ability as it was difficult for the participant to distinguish between tilt and no tilt when the tilt was low.

However, the ecological validity of the experiment is low, as these results are only applicable in a laboratory setting. Additionally, as there was only one participant, these results can not be generalised for any population. Thus, further research is required.

References

Onifade, M. (2021). Towards an emergency preparedness for self-rescue from underground coal mines. In Process Safety and Environmental Protection (pp. 946–957). Elsevier.

Sherman, B. S. (2024). Signal detection theory. EBSCO. https://www.ebsco.com/research-starters/health-and-medicine/signal-detection-theory

Wickens, C. D., & Flach, J. M. (1988). 5 - Information Processing. In Human Factors in Aviation (pp. 111–155). Academic Press.